6.4

2

(1) $\lim_{x\to\infty} \frac{\cosh x}{e^x}$

$= \lim_{x\to\infty} \frac{e^x + e^{-x}}{2e^x} = \lim_{x\to\infty} \frac{1 + e^{-2x}}{2} = \frac{1}{2}$ //

(3) $\lim_{x\to 0^+} \coth x$

$= \lim_{x\to 0^+} \frac{1}{\tanh x} = \lim_{x\to 0^+} \frac{\cosh x}{\sinh x} = \lim_{x\to 0^+} \frac{e^x + e^{-x}}{e^x - e^{-x}}$

$= \lim_{x\to 0^+} \frac{e^{2x} + 1}{e^{2x} - 1} = \infty$

3

(1) $y = \cosh e^x - e^{\cosh x}$

$y' = e^x \cdot \sinh e^x - \sinh x \cdot e^{\cosh x}$ //

5

(1) $\lim_{h\to 0} \frac{\sinh^{-1} h}{h}$

$= \lim_{h\to 0} \frac{\ln\left(h + \sqrt{h^2+1}\right)}{h}$

$= \lim_{h\to 0} \frac{1}{h+\sqrt{h^2+1}} \cdot \left(1 + \frac{2h}{2\sqrt{h^2+1}}\right) = 1$ //

9

(5)

$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = y$

$\frac{e^x - e^{-x}}{e^x + e^{-x}} = y$ 에서 x와 y 교환

$$\frac{e^{y}-e^{-y}}{e^{y}+e^{-y}} = x \quad \Leftrightarrow \quad \frac{e^{2y}-1}{e^{2y}+1} = x$$

$e^{y} = t$ 로 치환

$$\frac{t^{2}-1}{t^{2}+1} = x$$

$$t^{2}-1 = x(t^{2}+1)$$

$$(1-x)t^{2} = x+1$$

$$t^{2} = \frac{1+x}{1-x}$$

$$t = \sqrt{\frac{1+x}{1-x}}$$

$$e^{y} = \sqrt{\frac{1+x}{1-x}}$$

$$\therefore \quad y = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$